

Arunabha Majumder

Robotics & Mechatronics Researcher — Experimental Infrastructure, Sensing & Embedded Control

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Faculty of Engineering and Natural Sciences, Tampere University
Hervanta Campus, Tampere, Finland

Letter of Motivation: Staff Scientist, Autonomous and Electrified Mobile Work Machines

Dear Professor Ghabcheloo and members of the selection committee,

I am applying for the Staff Scientist position supporting research on autonomous and electrified mobile work machines. I am a robotics and mechatronics researcher completing my PhD in Mechatronics and Robotics at Aalborg University (expected defence: [month year]), with an MTech in Mechatronics and a BTech in Mechanical Engineering. Throughout my studies I have built and run experimental robotic systems, and the part of research I enjoy most is exactly what this role is about: making research hardware and software work reliably, so that ideas can be tested, demonstrated and taken to a higher readiness level.

Relevant skills and experience. In my PhD project VIEXO, I built a hybrid-actuated shoulder exoskeleton from concept to tested system: mechanical design in SolidWorks, 3D-printed prototypes, and integration of a 6 Nm motor, encoder, force sensors, data acquisition and embedded control. The design reduced energy consumption by more than 25% and received the Best Research Paper Award at IFToMM ISRM 2026. I also implemented an Assist-as-Needed controller on the hardware, and on a 5-bar planar parallel robot I developed a force-estimation method using variable-stiffness actuators and the robot Jacobian, validated against a force sensor to below 10% RMS error and used to measure human arm impedance in the gait lab. During my Master's at CSIR-CMERI, I built a pneumatic actuator test bench from raw materials, with valves, pressure sensors and DAQ, and developed a neural-network-based controller in MATLAB/Simulink. My software tools are Python, MATLAB/Simulink, LabVIEW and Arduino.

Laboratory practice, safety and teaching. I have designed and built custom automated test rigs and conducted experiments with 12 human participants using EMG and motion-capture equipment, which required careful experimental protocols, ethics approval and attention to participant safety. From 2023 to 2025 I taught university students and supervised Bachelor's students, which matches the thesis supervision part of this role. I also hold a Diploma in Fine Arts and create technical illustrations and visuals, which I would use for public demonstrators and public relations material.

What I would bring to the infrastructure. My experience is with laboratory robots and wearable systems rather than full-scale mobile work machines, and I would be keen to learn the heavy-machine platforms, their safety requirements and the software infrastructure from your team. I would focus on three things: (1) building well-documented, reusable test set-ups and sensor integrations, so that each research group does not start from zero; (2) helping researchers move from one-off experiments towards hardware and software that can be integrated into the shared infrastructure; and (3) clear safety procedures and documentation for the laboratory and test area. In the longer term, I want to grow into a staff scientist who connects research groups, students and industry partners around a well-maintained, first-class research infrastructure.

Why this matters to me. A strong research infrastructure multiplies the impact of every researcher who uses it. The combination of full-scale machines, a dedicated test area and close collaboration with industry in the Tampere ecosystem is a rare environment, and I would be glad to help maintain and develop it.

Referees: Professor Shaoping Bai, Department of Materials and Production, Aalborg University ([email]); [second referee name, title, institution, email].

Thank you for considering my application. I would welcome the opportunity to discuss how I can contribute.

Kind regards,
Arunabha Majumder